Application Logs

Using Log4J

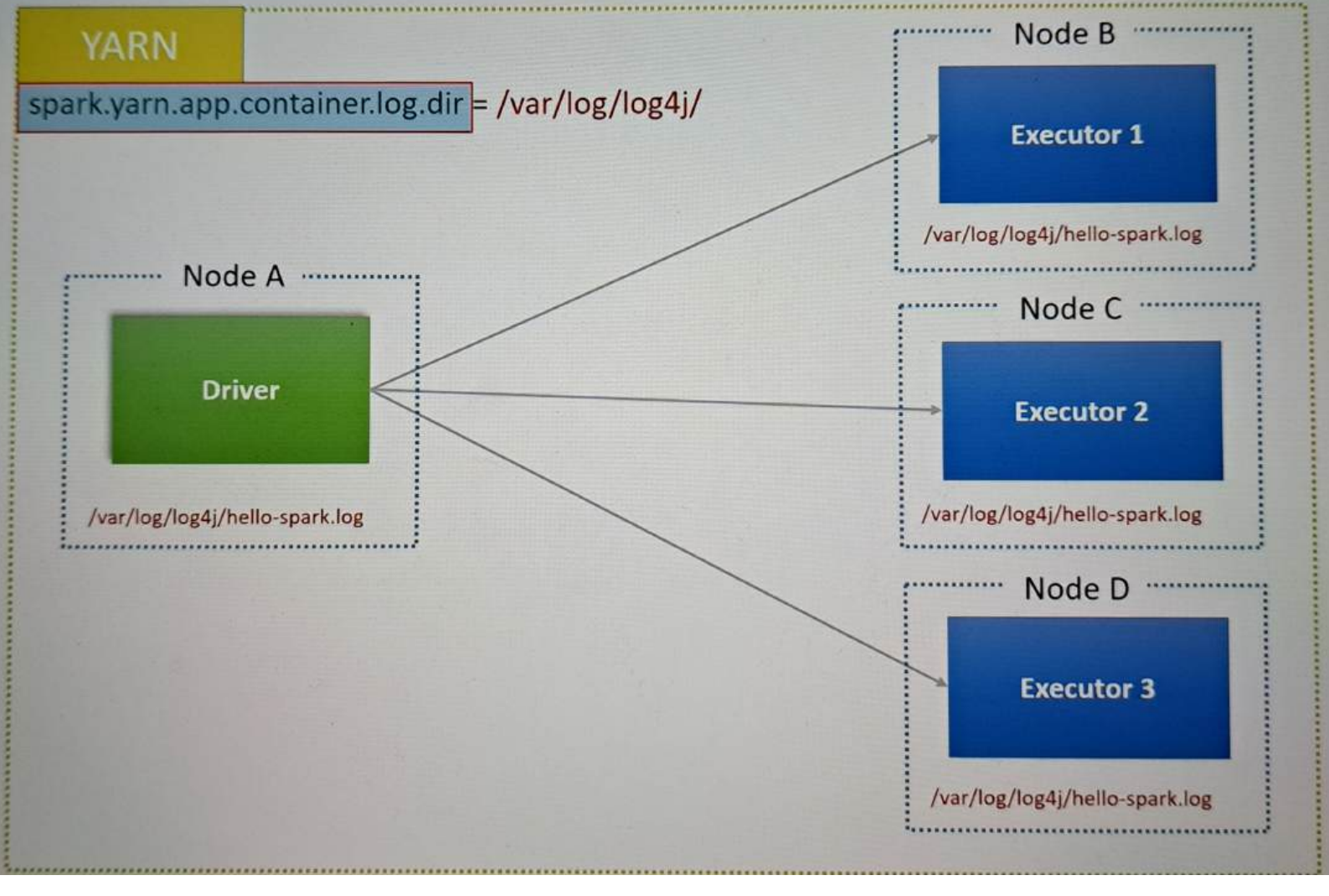
1. Create a Log4J configuration file
2. Configure Spark JVM to pickup the Log4J configuration file
3. Create a Python Class to get Spark's Log4J instance and use it

```
# Set everything to be logged to the console
log4j.rootCategory=WARN, console

# define console appender
log4j.appender.console=org.apache.log4j.ConsoleAppender
log4j.appender.console.target=System.out
log4j.appender.console.layout=org.apache.log4j.PatternLayout
log4j.appender.console.layout.ConversionPattern=%d{yy/MM/dd HH:mm:ss} %p %c{1}: %m%n

#application log
log4j.logger.guru.learningjournal.spark.examples=INFO, console, file
log4j.additivity.guru.learningjournal.spark.examples=false

#define rolling file appender
log4j.appender.file=org.apache.log4j.RollingFileAppender
log4j.appender.file.File=${spark.yarn.app.container.log.dir}/${logfile.name}.log
#define following in Java System
# -Dlog4j.configuration=file:log4j.properties
# -Dlogfile.name=hello-spark
# -Dspark.yarn.app.container.log.dir=app-logs
log4j.appender.file.ImmediateFlush=true
log4j.appender.file.Append=false
log4j.appender.file.MaxFileSize=500MB
log4j.appender.file.MaxBackupIndex=2
log4j.appender.file.layout=org.apache.log4j.PatternLayout
log4j.appender.file.layout.conversionPattern=%d{yy/MM/dd HH:mm:ss} %p %c{1}: %m%n

# Recommendations from Spark template
```

Using Log4J

1. Logger
2. Configurations
3. Appender

Application Logs



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How to configure JVM variables?

udemy

Application Logs

Setting JVM Parameters

SPARK_HOME/conf/spark-defaults.conf

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C:\Windows\System32\cmd.exe

```
Microsoft Windows [Version 10.0.18362.836]
(c) 2019 Microsoft Corporation. All rights reserved.

C:\demo\spark245>echo %SPARK_HOME%
C:\demo\spark245

C:\demo\spark245>
```

conf

File Home Share View

This PC > Windows7_OS (C:) > demo > spark245 > conf

Quick access
OneDrive - Personal
Documents
Email attachments
Music
Pictures
Prashant
Sanchit
This PC
3D Objects
Desktop
Documents
Downloads
Music
Pictures
Videos
Windows7_OS (C:)
Backup (E:)
Network

Name	Date modified	Type	Size
docker.properties.template	2/3/2020 1:17 AM	TEMPLATE File	1 KB
fairscheduler.xml.template	2/3/2020 1:17 AM	TEMPLATE File	2 KB
log4j.properties.archived	5/31/2020 1:06 PM	ARCHIVED File	2 KB
log4j.properties.template	2/3/2020 1:17 AM	TEMPLATE File	2 KB
metrics.properties.template	2/3/2020 1:17 AM	TEMPLATE File	8 KB
slaves.template	2/3/2020 1:17 AM	TEMPLATE File	1 KB
spark-defaults.conf	5/31/2020 1:06 PM	CONF File	2 KB
spark-defaults.conf.template	2/3/2020 1:17 AM	TEMPLATE File	2 KB
spark-env.sh.template	2/3/2020 1:17 AM	TEMPLATE File	5 KB

*C:\demo\spark245\conf\spark-defaults.conf - Notepad++

File Edit Search View Encoding Language Settings Macro Run Plugins Window ?

spark-defaults.conf

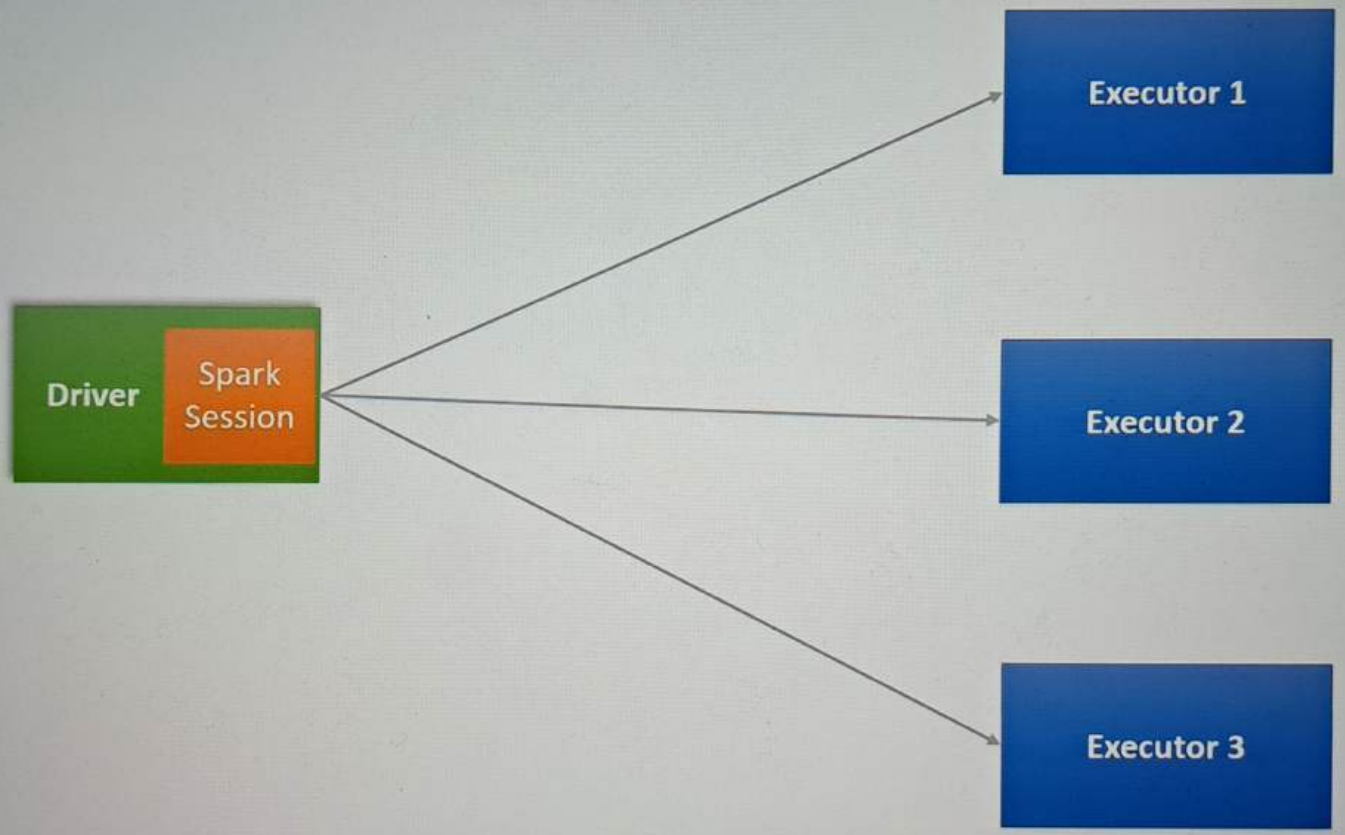
```
#
# Licensed to the Apache Software Foundation (ASF) under one or more
# contributor license agreements.  See the NOTICE file distributed with
# this work for additional information regarding copyright ownership.
# The ASF licenses this file to You under the Apache License, Version 2.0
# (the "License"); you may not use this file except in compliance with
# the License.  You may obtain a copy of the License at
#
#    http://www.apache.org/licenses/LICENSE-2.0
#
# Unless required by applicable law or agreed to in writing, software
# distributed under the License is distributed on an "AS IS" BASIS,
# WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
# See the License for the specific language governing permissions and
# limitations under the License.
#

# Default system properties included when running spark-submit.
# This is useful for setting default environmental settings.

# Example:
# spark.master                     spark://master:7077
# spark.eventLog.enabled           true
# spark.eventLog.dir               hdfs://namenode:8021/directory
# spark.serializer                 org.apache.spark.serializer.KryoSerializer
# spark.driver.memory              5g
# spark.executor.extraJavaOptions  -XX:+PrintGCDetails -Dkey=value -Dnumbers="one two three"
spark.driver.extraJavaOptions      -Dlog4j.configuration=file:log4j.properties -Dspark.yarn.app.container.log.dir=app-logs -Dlogfile.name=hello-spark
```

Normal text file | length : 1,442 lines : 28 | Ln : 28 Col : 1 Sel : 0 | 0 | Unix (LF) | UTF-8 | INS

Spark Session



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Anaconda Prompt (Anaconda3) - bin\pyspark

```
(base) C:\Users\prashant>cd C:\demo\spark245

(base) C:\demo\spark245>bin\pyspark
Python 3.7.6 (default, Jan  8 2020, 20:23:39) [MSC v.1916 64 bit (AMD64)] :: Anaconda, Inc. on win32

Warning:
This Python interpreter is in a conda environment, but the environment has
not been activated.  Libraries may fail to load.  To activate this environment
please see https://conda.io/activation

Type "help", "copyright", "credits" or "license" for more information.
20/06/07 15:05:28 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... us
ing builtin-java classes where applicable
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /__ / .__/\_,_/_/ /_/\_\   version 2.4.5
      /_/

Using Python version 3.7.6 (default, Jan  8 2020 20:23:39)
SparkSession available as 'spark'.
>>>
```

File Edit View Navigate Code Refactor Run Tools VCS Window Help HelloSpark - HelloSpark.py - PyCharm

HelloSpark > HelloSpark.py

Project

- HelloSpark C:\Users\prashant\Documents\GitHub\Spark-Program
  - lib
    - __init__.py
    - logger.py
  - HelloSpark.py
  - log4j.properties
- External Libraries
- Scratches and Consoles

HelloSpark.py | logger.py | log4j.properties

```python
from pyspark.sql import *

from lib.logger import Log4J

if __name__ == "__main__":
    spark = SparkSession().builder \
        .appName("Hello Spark") \
        .master("local[3]") \
        .getOrCreate()

    logger = Log4J(spark)

    logger.info("Starting HelloSpark")
    # Your processing code

    logger.info("Finished HelloSpark")
    spark.stop()
```

HelloSpark.py × logger.py × log4j.properties ×

```python
class Log4J:
    def __init__(self, spark):
        log4j = spark._jvm.org.apache.log4j
        root_class = "guru.learningjournal.spark.examples"
        conf = spark.sparkContext.getConf()
        app_name = conf.get("spark.app.name")
        self.logger = log4j.LogManager.getLogger(root_class + "." + "HelloSpark")

    def warn(self, message):
        self.logger.warn(message)

    def info(self, message):
        self.logger.info(message)

    def error(self, message):
        self.logger.error(message)

    def debug(self, message):
        self.logger.debug(message)
```

File Edit View Navigate Code Refactor Run Tools VCS Window Help HelloSpark - HelloSpark.py - PyCharm

HelloSpark > HelloSpark.py

Project

HelloSpark C:\Users\prashant\Documents\GitHub\Spark-Program
- lib
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HelloSpark.py | logger.py | log4j.properties

```python
from pyspark.sql import *

from lib.logger import Log4J

if __name__ == "__main__":
    spark = SparkSession.builder \
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        .getOrCreate()

    logger = Log4J(spark)

    logger.info("Starting HelloSpark")
    # Your processing code

    logger.info("Finished HelloSpark")
    spark.stop()
```

if __name__ == "__main__"

Run: HelloSpark

```
Traceback (most recent call last):
  File "C:\Users\prashant\anaconda3\envs\HelloSpark\lib\site-packages\py4j\java_gateway.py", line 985, in send_command
    response = connection.send_command(command)
  File "C:\Users\prashant\anaconda3\envs\HelloSpark\lib\site-packages\py4j\java_gateway.py", line 1164, in send_command
    "Error while receiving", e, proto.ERROR_ON_RECEIVE)
py4j.protocol.Py4JNetworkError: Error while receiving

Process finished with exit code 0
```

9: Git | 6: TODO | 4: Run | Terminal | Python Console | Event Log

6:25 CRLF UTF-8 4 spaces Python 3.7 (HelloSpark) master

File Edit View Navigate Code Refactor Run Tools VCS Window Help HelloSpark - hello-spark.log - PyCharm

HelloSpark 〉 app-logs 〉 hello-spark.log

Project

HelloSpark C:\Users\prashant\Documents\GitHub\Spark-Program
- app-logs
  - hello-spark.log
- lib
  - __init__.py
  - logger.py
- HelloSpark.py
- log4j.properties
- External Libraries
- Scratches and Consoles

HelloSpark.py | hello-spark.log | logger.py | log4j.properties

```
20/06/07 16:48:04 INFO Hello Spark: Starting HelloSpark
20/06/07 16:48:04 INFO Hello Spark: Finished HelloSpark
```

Run: HelloSpark

```
C:\Users\prashant\anaconda3\envs\HelloSpark\python.exe C:/Users/prashant/Documents/GitHub/Spark-Programming-In-Python/HelloSpark/HelloSpark.py
20/06/07 16:48:01 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
20/06/07 16:48:04 INFO Hello Spark: Starting HelloSpark
20/06/07 16:48:04 INFO Hello Spark: Finished HelloSpark

Process finished with exit code 0
```

Spark Session Configs

Configuring Spark Session

1. Environment variables
2. SPARK_HOME\conf\spark-defaults.conf
3. spark-submit command-line options
4. SparkConf Object

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Spark Session Configs

C:\Windows\System32\cmd.exe

```
Microsoft Windows [Version 10.0.18362.836]
(c) 2019 Microsoft Corporation. All rights reserved.

C:\demo\spark245>bin\spark-submit --master local[3] --conf "spark.app.name=Hello Spark" --conf spark.eventLog.enabled=false HelloSpark.py
```

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HelloSpark.py

```python
from pyspark.sql import *

from lib.logger import Log4J

if __name__ == "__main__":
    spark = SparkSession.builder \
        .appName("Hello Spark") \
        .master("local[3]") \
        .getOrCreate()

    logger = Log4J(spark)

    logger.info("Starting HelloSpark")
    # Your processing code

    logger.info("Finished HelloSpark")
    # spark.stop()
```

HelloSpark.py

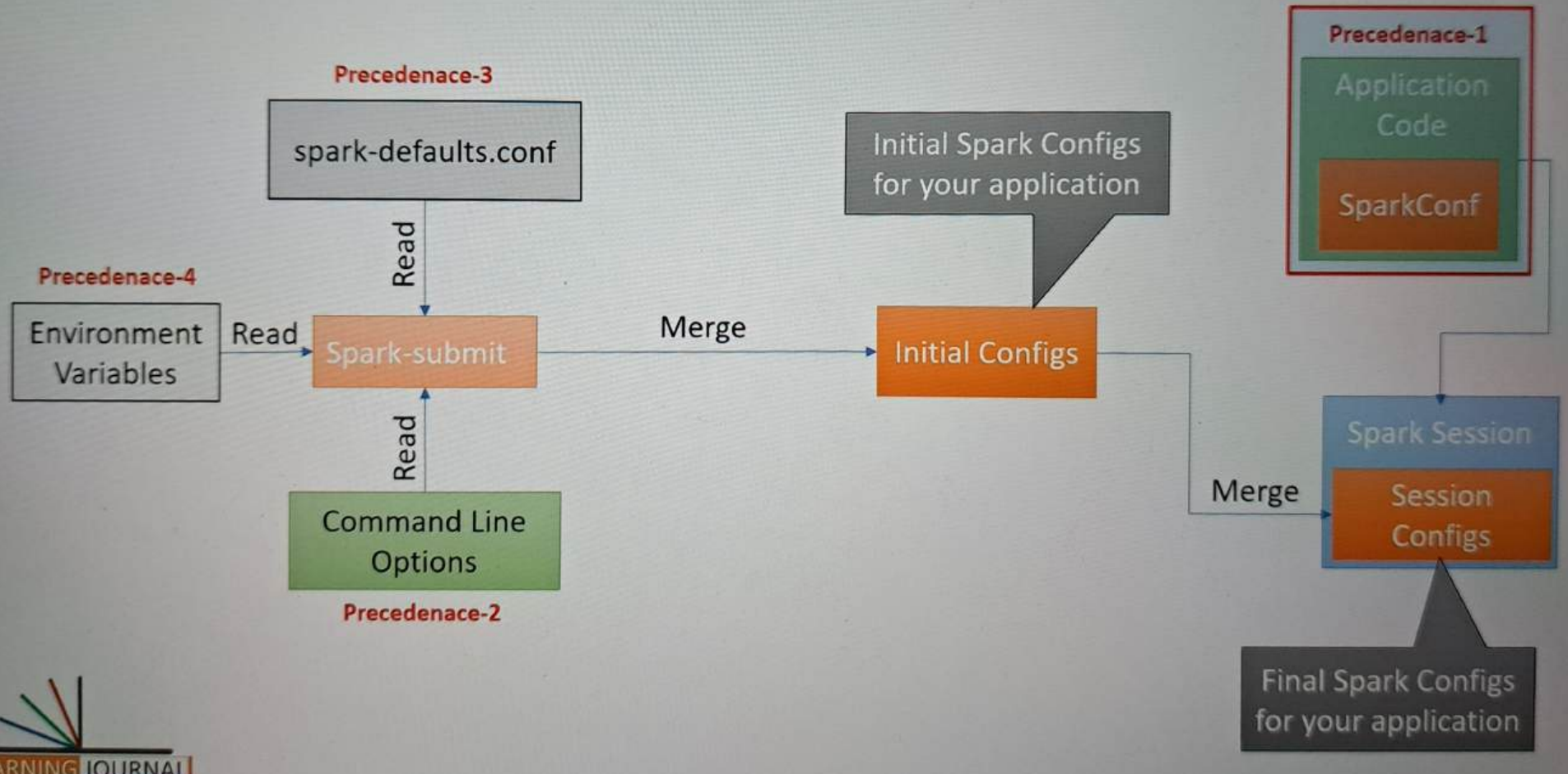
```python
from pyspark import SparkConf
from pyspark.sql import *

from lib.logger import Log4J

if __name__ == "__main__":
    conf = SparkConf()
    conf.set("spark.app.name", "Hello Spark")
    conf.set("spark.master", "local[3]")
    spark = SparkSession.builder \
        .appName("Hello Spark") \
        .master("local[3]") \
        .getOrCreate()

    logger = Log4J(spark)

    logger.info("Starting HelloSpark")
    # Your processing code

    logger.info("Finished HelloSpark")
    # spark.stop()
```

https://spark.apache.org/docs/latest/configuration.html#application-properties

HelloSpark.py

```python
from pyspark import SparkConf
from pyspark.sql import *

from lib.logger import Log4J

if __name__ == "__main__":
    conf = SparkConf()
    conf.set("spark.app.name", "Hello Spark")
    conf.set("spark.master", "local[3]")
    spark = SparkSession.builder \
        .config(conf=conf) \
        .getOrCreate()

    logger = Log4J(spark)

    logger.info("Starting HelloSpark")
    # Your processing code

    logger.info("Finished HelloSpark")
    # spark.stop()
```

Spark Session Configs



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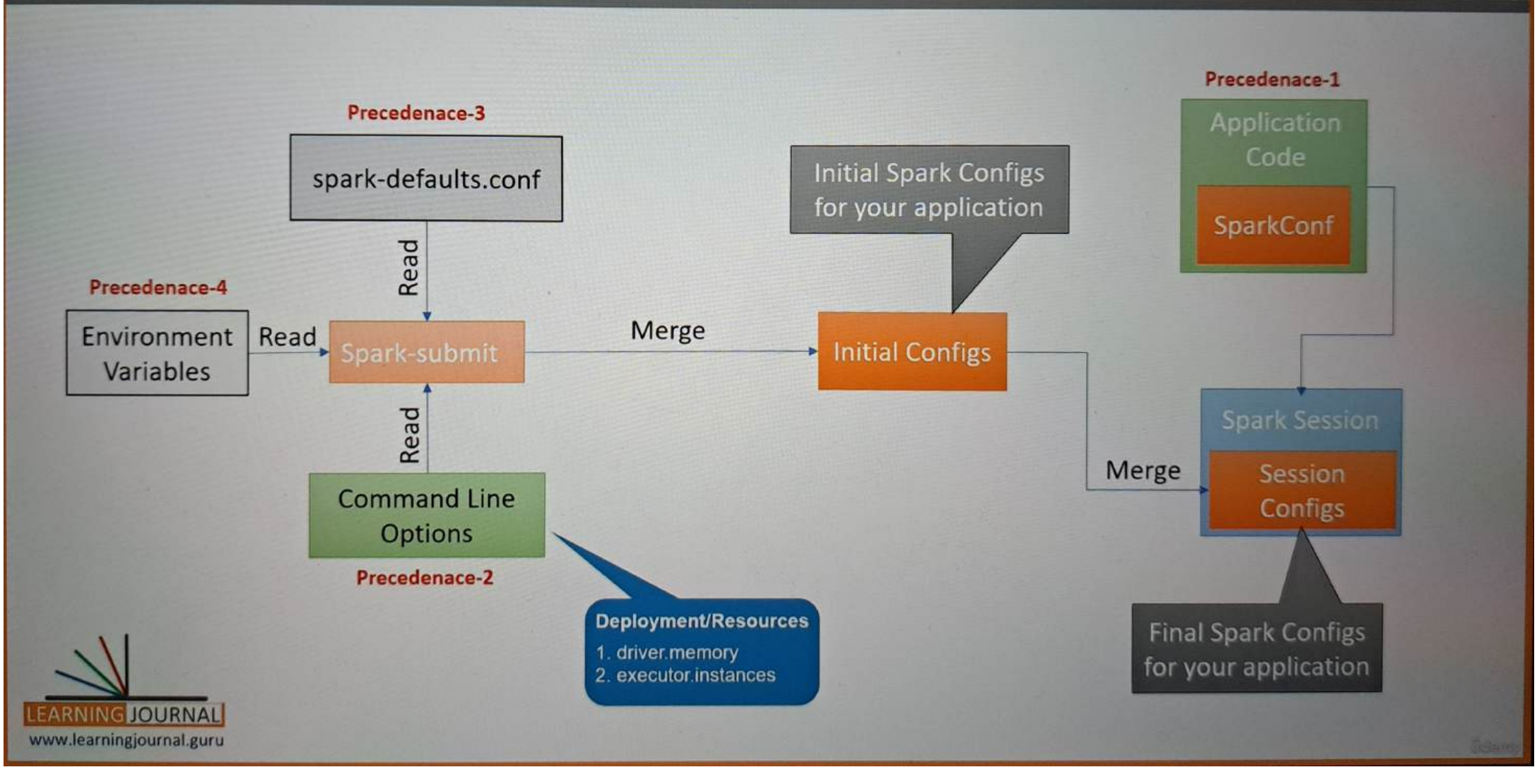
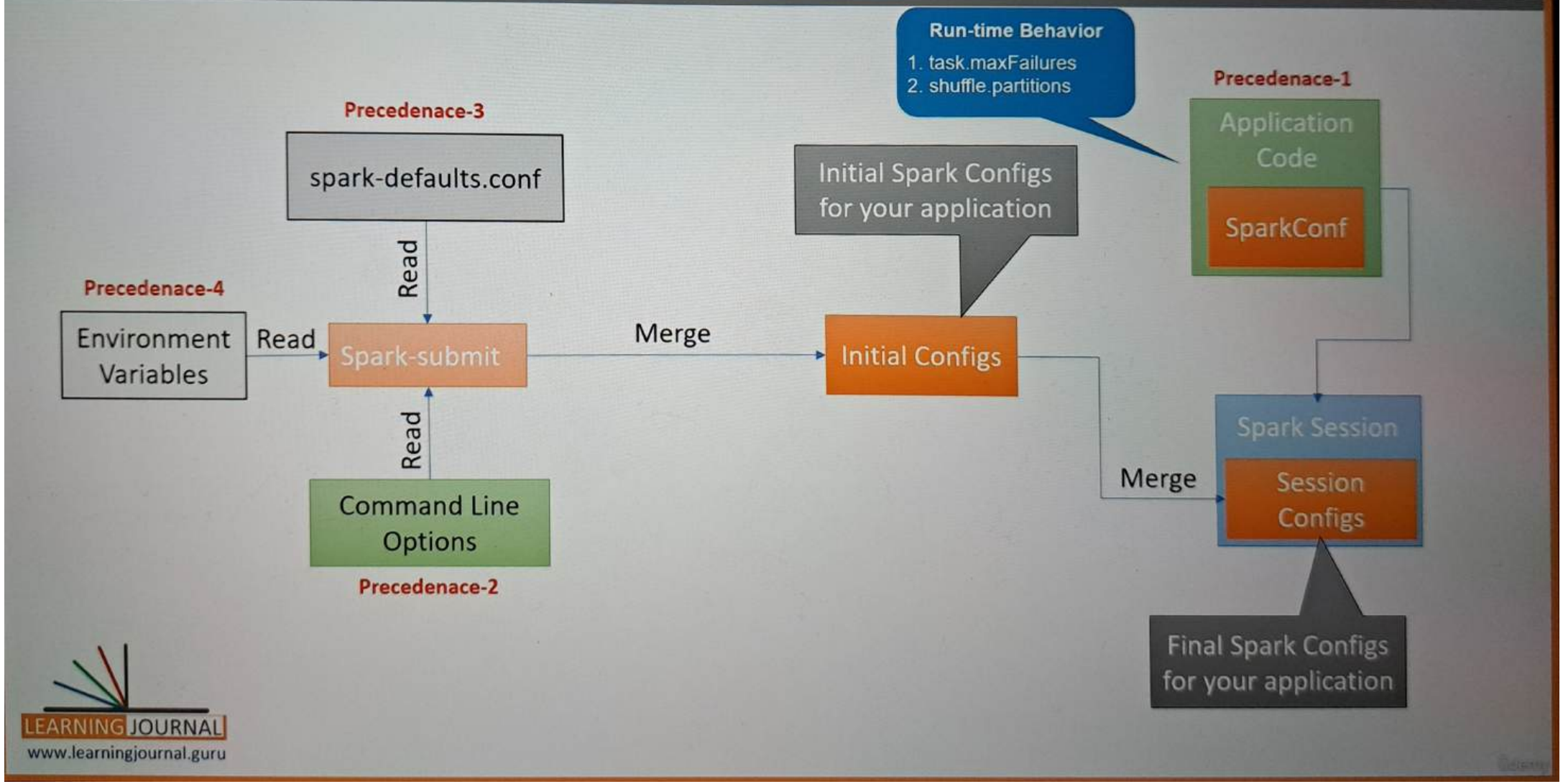
Spark Session Configs



Spark Session Configs



File Edit View Navigate Code Refactor Run Tools VCS Window Help HelloSpark - utils.py - PyCharm

HelloSpark > lib > utils.py

Project

- HelloSpark C:\Users\prashant\Documents\GitHub\Spark-Programmi
  - app-logs
  - lib
    - __init__.py
    - logger.py
    - utils.py
  - HelloSpark.py
  - log4j.properties
  - spark.conf
- External Libraries
- Scratches and Consoles

HelloSpark.py | utils.py

```python
import configparser

from pyspark import SparkConf


def get_spark_app_config():
    spark_conf = SparkConf()
    config = configparser.ConfigParser()
    config.read("spark.conf")

    for (key, val) in config.items("SPARK_APP_CONFIGS"):
        spark_conf.set(key, val)
    return spark_conf

```

File Edit View Navigate Code Refactor Run Tools VCS Window Help HelloSpark - spark.conf - PyCharm

HelloSpark 〉 spark.conf

Project

- HelloSpark C:\Users\prashant\Documents\GitHub\Spark-Programmi
  - app-logs
  - lib
  - HelloSpark.py
  - log4j.properties
  - spark.conf
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- Scratches and Consoles

HelloSpark.py | spark.conf

```
[SPARK_APP_CONFIGS]
spark.app.name = Hello Spark
spark.master = local[3]
```

File Edit View Navigate Code Refactor Run Tools VCS Window Help HelloSpark - HelloSpark.py - PyCharm

HelloSpark > HelloSpark.py

Project

- HelloSpark C:\Users\prashant\Documents\GitHub\Spark-Programmi
  - app-logs
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    - __init__.py
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    - utils.py
  - HelloSpark.py
  - log4j.properties
  - spark.conf
- External Libraries
- Scratches and consoles

HelloSpark.py | utils.py

```python
from pyspark import SparkConf
from pyspark.sql import *

from lib.logger import Log4J
from lib.utils import get_spark_app_config

if __name__ == "__main__":
    conf = get_spark_app_config()
    spark = SparkSession.builder \
        .config(conf=conf) \
        .getOrCreate()

    logger = Log4J(spark)

    logger.info("Starting HelloSpark")
    # Your processing code

    logger.info("Finished HelloSpark")
    # spark.stop()
```

File Edit View Navigate Code Refactor Run Tools VCS Window Help HelloSpark - HelloSpark.py - PyCharm

HelloSpark 〉 HelloSpark.py

Project

HelloSpark C:\Users\prashant\Documents\GitHub\Spark-Programmi
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- HelloSpark.py
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- spark.conf
- External Libraries
- Scratches and Consoles

HelloSpark.py | spark.conf | utils.py

```python
from pyspark import SparkConf
from pyspark.sql import *

from lib.logger import Log4J
from lib.utils import get_spark_app_config

if __name__ == "__main__":
    conf = get_spark_app_config()
    spark = SparkSession.builder \
        .config(conf=conf) \
        .getOrCreate()

    logger = Log4J(spark)

    logger.info("Starting HelloSpark")
    # Your processing code
    conf_out = spark.sparkContext.getConf()
    logger.info(conf_out.toDebugString())
```

if __name__ == "__main__"

Run: HelloSpark

```
C:\Users\prashant\anaconda3\envs\HelloSpark\python.exe C:/Users/prashant/Documents/GitHub/Spark-Programming-In-Python/HelloSpark/HelloSpark.py
```

9: Git | 6: TODO | 4: Run | Terminal | Python Console | Event Log

19:5 CRLF UTF-8 4 spaces Python 3.7 (HelloSpark) master